

Table of contents

SIR Epidemic on a Graph

Simon Frost 2026-03-29

- Introduction
- Setup
- Create a graph
- Individual-based model (order-1)
- Pair-based model (order-2)
- Gillespie validation
- Discussion

Introduction

In contrast to the edge-based approach of [EdgeBasedModels.jl](#), which works with degree distributions and probability generating functions, `NodeBasedModels.jl` operates directly on specific graph instances. This means every node and every edge is tracked explicitly, allowing us to capture the full heterogeneity of a network — not just its degree distribution, but the precise adjacency structure.

[Sharkey \(2011\)](#) showed that node-level epidemic models can be organised into a hierarchy of moment closures:

- Order-1 (individual-based / NIMFA): Tracks the marginal probability $\langle S_i \rangle, \langle I_i \rangle$ for each node i . Pairs are approximated by independence:

$$\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle.$$

- Order-2 (pair-based): Tracks both node marginals and pair probabilities $\langle S_i I_j \rangle$ for each edge (i, j) . Triples are closed via the Kirkwood superposition:

$$\langle A_k B_i C_j \rangle \approx \frac{\langle A_k B_i \rangle \langle B_i C_j \rangle}{\langle B_i \rangle}.$$

- Exact (Gillespie): Continuous-time Markov chain simulation using the Gillespie algorithm. No approximation — the “gold standard.”

Higher-order closures are more accurate but more expensive. In this vignette we run all three levels on the same graph and compare their predictions for an SIR epidemic.

Setup

```
using NodeBasedModels
using Graphs
using Plots
using OrdinaryDiffEqDefault
using Random
```

Create a graph

We build a random regular graph with $N = 100$ nodes and degree $k = 6$. Every node has exactly 6 neighbours.

```
Random.seed!(42)
g = random_regular_graph(100, 6)
net = GraphNetwork(g)

println("Nodes:      ", nv(g))
println("Edges:      ", ne(g))
println("Mean degree: ", mean_degree(net))
```

```
Nodes:      100
Edges:      300
Mean degree: 6.0
```

Individual-based model (order-1)

The individual-based (NIMFA) approximation tracks the probability that each node is in a given state. For the SIR model on a graph with adjacency matrix A , the equations for node i are

$$\frac{d\langle S_i \rangle}{dt} = -\tau \sum_j A_{ij} \langle S_i \rangle \langle I_j \rangle, \quad \frac{d\langle I_i \rangle}{dt} = \tau \sum_j A_{ij} \langle S_i \rangle \langle I_j \rangle - \gamma \langle I_i \rangle.$$

Because pairs are factorised ($\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle$), the system has $2N$ ODEs.